PROBLEM NS17

Rotating Porous Disk with Suction

Original statement · NS17.pdf

Similarity solution | Rotating disk with suction | Boundary conditions | Pressure distribution

OFFICIAL PROBLEM STATEMENT

We will consider an infinitely large porous disk rotating with angular velocity Ω that sucks the fluid with a known velocity $v_z = -2(\nu\Omega)^{1/2}\lambda_1$. The velocity induced in the fluid of density ρ and kinematic viscosity $\nu = \mu/\rho$ is three-dimensional, with components (v_r, v_θ, v_z) in the cylindrical coordinates system shown in the sketch.

We want to show that the solution of the problem has the form $v_z = -2(\nu\Omega)^{1/2} f(\eta)$, $v_\theta = r\Omega g(\eta)$, where $\eta = (\Omega/\nu)^{1/2} z$ is a similarity variable and $f(\eta)$ and $g(\eta)$ are two functions that must be obtained in the course of the solution, such that the solution far from the disk, as $\eta \to \infty$, is given by $v_r \to 0, v_\theta \to 0$.

Taking into account the azimuthal symmetry of the problem,

1. Use the continuity equation to obtain the radial component v_r in terms of the function $f(\eta)$.

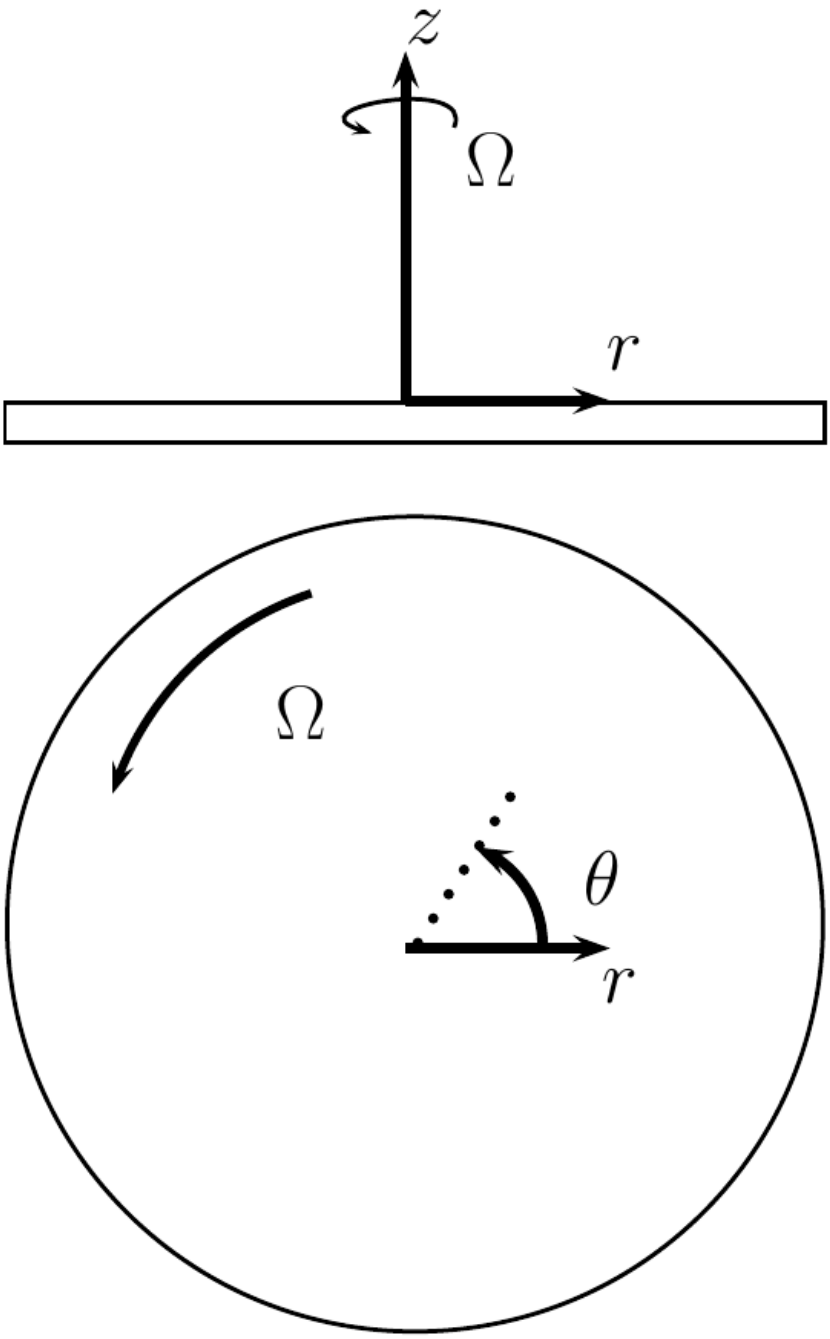
2. Use the azimuthal momentum equation to find the ordinary differential equation to be satisfied by the function $g(\eta)$.

3. Use the physical boundary conditions for $\bar{v}$ to find the boundary conditions that $f(\eta)$ and $g(\eta)$ must satisfy.

4. Use the vertical momentum equation to obtain the pressure distribution in terms of an integration function $C(r)$ that will appear in the course of the integration.

5. Use the radial momentum equation to determine $C(r)$ and the ordinary differential equation to be satisfied by $f(\eta)$, with its corresponding boundary conditions. Use the variable separation method to show that $C(r) = A$, where A is a constant. Obtain A in terms of the pressure far from the plate p_∞ and the pressure distribution $p(\eta)$.

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Statement figure (crop of NS17.pdf, p. 1): porous disk rotating with angular velocity Ω, side view (axes z, r) and top view (polar angle θ)

Source: NS17.pdf (page 1 of 2: statement; page 2: handwritten solution). Notes.pdf, Chapter 4: continuity Eq. 4.40 and momentum Eq. 4.41 in cylindrical coordinates.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, axisymmetric ($\partial/\partial\theta = 0, \partial/\partial t = 0$) flow of an incompressible liquid with constant ρ and ν above an infinitely large rotating porous disk, without mass forces. Equations (4.40) and (4.41) in cylindrical coordinates.
- The disk (plane $z = 0$, the liquid occupies $z > 0$) rotates with the angular velocity Ω and sucks the liquid with the known axial velocity $v_z(0) = -2(\nu\Omega)^{1/2}\lambda_1$, with λ_1 a given constant.
- Self-similar ansatz of the statement: $v_z = -2(\nu\Omega)^{1/2} f(\eta)$, $v_\theta = r\Omega\, g(\eta)$, with $\eta = (\Omega/\nu)^{1/2} z$; v_r is to be obtained.
- Far from the disk ($\eta \to \infty$): $v_r \to 0$, $v_\theta \to 0$ and the pressure tends to p_∞.
- Data: $[\rho,\ \nu,\ \Omega,\ \lambda_1,\ p_\infty]$. Unknowns: v_r, the ODEs for f and g with their boundary conditions, $p(r, z)$ and the constant A.
- Derivatives used: $d\eta/dz = (\Omega/\nu)^{1/2}$, so that $\partial/\partial z = (\Omega/\nu)^{1/2}\, d/d\eta$.

Symbol	Meaning	SI unit
Ω	angular velocity of the disk	rad/s
$\nu = \mu/\rho$	kinematic viscosity	m^2/s
$\eta = (\Omega/\nu)^{1/2} z$	similarity variable	1
$f(\eta), g(\eta)$	dimensionless axial and azimuthal functions	1
λ_1	dimensionless suction parameter	1
p_∞	pressure far from the disk	Pa

PHASE 2

Frames, Field Form and Boundary Conditions

- Cylindrical frame (r, θ, z) with z along the axis of rotation, normal to the disk and pointing into the liquid; the basis $\{\bar{e}_r, \bar{e}_\theta, \bar{e}_z\}$ is orthonormal and right-handed (see the sketch of the statement).
- Field form: $v_r = v_r(r, z)$, $v_\theta = r\Omega\, g(\eta)$, $v_z = -2(\nu\Omega)^{1/2} f(\eta)$, $p = p(r, z)$.

Location	Condition	Origin
$z = 0$	$v_r = 0, v_\theta = r\Omega$	no-slip on the rotating disk
$z = 0$	$v_z = -2(\nu\Omega)^{1/2}\lambda_1$	known suction velocity
$z \to \infty$	$v_r \to 0, v_\theta \to 0$	liquid at rest far from the disk
$z \to \infty$	$p \to p_\infty$	pressure far from the disk

PHASE 3

Step-by-Step Derivation

3.1 Radial velocity from continuity (item 1)

$$\frac{1}{r}\frac{\partial (r v_r)}{\partial r} + \underbrace{\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}}_{\partial_\theta = 0} + \frac{\partial v_z}{\partial z} = 0 \tag{1}$$

The axial derivative is, by the chain rule, $\dfrac{\partial v_z}{\partial z} = \dfrac{dv_z}{d\eta}\dfrac{d\eta}{dz} = -2(\nu\Omega)^{1/2} f' \left(\tfrac{\Omega}{\nu}\right)^{1/2} = -2\Omega f'(\eta)$. Hence $\dfrac{1}{r}\dfrac{\partial (r v_r)}{\partial r} = 2\Omega f'$. Multiplying by r and integrating in r at fixed z (so f' is constant): $r\, v_r = \Omega f'\, r^2 + F(z)$; the radial velocity must be finite on the axis $r = 0$, which requires $F(z) = 0$:

$$\boxed{v_r = r\,\Omega\, f'(\eta)} \tag{2}$$

3.2 Azimuthal momentum (item 2)

The θ component of (4.41), steady, with $\partial/\partial\theta = 0$ and no mass force:

$$\rho\left(v_r\frac{\partial v_\theta}{\partial r} + \frac{v_r v_\theta}{r} + v_z\frac{\partial v_\theta}{\partial z}\right) = \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial (r v_\theta)}{\partial r}\right) + \frac{\partial^2 v_\theta}{\partial z^2}\right] \tag{3}$$

Each term for $v_\theta = r\Omega g, v_r = r\Omega f', v_z = -2(\nu\Omega)^{1/2} f$ (the η derivatives bring $(\Omega/\nu)^{1/2}$):

- $v_r\, \partial v_\theta/\partial r = (r\Omega f')(\Omega g) = r\Omega^2 f' g$ and $v_r v_\theta / r = (r\Omega f')(\Omega g) = r\Omega^2 f' g$;
- $v_z\, \partial v_\theta/\partial z = -2(\nu\Omega)^{1/2} f \cdot r\Omega g' \,(\Omega/\nu)^{1/2} = -2r\Omega^2 f g'$;

- the first viscous term is zero: $r^{-1}\partial(rv_\theta)/\partial r = r^{-1}\partial(r^2\Omega g)/\partial r = 2\Omega g$ does not depend on r;
- $\partial^2 v_\theta/\partial z^2 = r\Omega g''\ \Omega/\nu$.

Substituting and using $\mu/\nu = \rho$:

$$\rho\, r\Omega^2\,(2f'g - 2fg') = \rho\, r\Omega^2 g'' \implies \boxed{g'' + 2\,(f\,g' - f'g) = 0} \tag{4}$$

3.3 Boundary conditions (item 3)

At the disk, $z = 0$ ($\eta = 0$): $v_r = r\Omega f'(0) = 0$ gives $f'(0) = 0$; $v_\theta = r\Omega g(0) = r\Omega$ gives $g(0) = 1$; $v_z = -2(\nu\Omega)^{1/2} f(0) = -2(\nu\Omega)^{1/2}\lambda_1$ gives $f(0) = \lambda_1$. Far from the disk ($\eta \to \infty$): $v_r \to 0$ gives $f'(\infty) = 0$ and $v_\theta \to 0$ gives $g(\infty) = 0$:

$$\boxed{f(0) = \lambda_1, \quad f'(0) = 0, \quad f'(\infty) = 0, \qquad g(0) = 1, \quad g(\infty) = 0} \tag{5}$$

3.4 Vertical momentum and the pressure (item 4)

The z component of (4.41): $v_r\,\partial v_z/\partial r = 0$ because v_z depends only on z, and the viscous term is $\mu\,\partial^2 v_z/\partial z^2$ (the radial part $r^{-1}\partial(r\partial v_z/\partial r)/\partial r$ vanishes):

$$\rho\, v_z \frac{\partial v_z}{\partial z} = -\frac{\partial p}{\partial z} + \mu\frac{\partial^2 v_z}{\partial z^2} \implies \frac{\partial}{\partial z}\left[p + \frac{\rho v_z^2}{2} - \mu\frac{\partial v_z}{\partial z}\right] = 0 \tag{6}$$

Integrating in z at fixed r, the bracket is a function $C(r)$:

$$p + \frac{\rho v_z^2}{2} - \mu\frac{\partial v_z}{\partial z} = C(r) \tag{7}$$

With $\rho v_z^2/2 = 2\rho\nu\Omega f^2 = 2\mu\Omega f^2$ and $-\mu\,\partial v_z/\partial z = 2\mu\Omega f'$:

$$\boxed{p(r,\eta) = C(r) - 2\mu\Omega\left[f^2(\eta) + f'(\eta)\right]} \tag{8}$$

3.5 Radial momentum, the ODE for f and the constant A (item 5)

The r component of (4.41) with the viscous terms $\mu\left[\partial\left(r^{-1}\partial(rv_r)/\partial r\right)/\partial r + \partial^2 v_r/\partial z^2\right]$:

$$\rho\left(v_r\frac{\partial v_r}{\partial r} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial p}{\partial r} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial(rv_r)}{\partial r}\right) + \frac{\partial^2 v_r}{\partial z^2}\right] \tag{9}$$

Each term: $v_r\,\partial v_r/\partial r = (r\Omega f')(\Omega f') = r\Omega^2 f'^2$; $v_\theta^2/r = r\Omega^2 g^2$; $v_z\,\partial v_r/\partial z = -2(\nu\Omega)^{1/2} f\cdot r\Omega f''(\Omega/\nu)^{1/2} = -2r\Omega^2 f f''$; the first viscous term is zero because $r^{-1}\partial(r^2\Omega f')/\partial r = 2\Omega f'$ does not depend on r; $\partial^2 v_r/\partial z^2 = r\Omega f'''\,\Omega/\nu$. From (8), $\partial p/\partial r = C'(r)$, since the η term does not depend on r. Substituting and using $\mu/\nu = \rho$:

$$\rho r\Omega^2\left(f'^2 - g^2 - 2ff''\right) = -C'(r) + \rho r\Omega^2 f''' \implies -f''' + f'^2 - g^2 - 2ff'' = -\frac{C'(r)}{\rho r\Omega^2} \tag{10}$$

The left side depends only on η and the right side only on r: both equal a constant, $-\lambda$. As $\eta\to\infty$, $f'\to 0$ (and so $f''\to 0$, $f'''\to 0$) and $g\to 0$: the left side tends to 0, so $\lambda = 0$. Therefore $C'(r) = 0$, that is

$$\boxed{C(r) = A\ (\text{a constant}), \qquad f''' + 2ff'' - f'^2 + g^2 = 0} \tag{11}$$

with the boundary conditions (5) for f ($f(0) = \lambda_1$, $f'(0) = f'(\infty) = 0$); the system (4)-(11) for f and g, with the five conditions (5), is a nonlinear boundary-value problem.

The constant A follows from (8) as $\eta\to\infty$ ($p\to p_\infty$, $f'\to 0$, $f\to f(\infty)$):

$$p_\infty = A - 2\mu\Omega f(\infty)^2 \implies \boxed{A = p_\infty + 2\mu\Omega\, f(\infty)^2} \tag{12}$$

and the pressure distribution is

$$\boxed{p(\eta) = p_\infty - 2\mu\Omega\left[f^2(\eta) - f^2(\infty) + f'(\eta)\right]} \tag{13}$$

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The disk drags the liquid into rotation and the centrifugal force throws it outwards (the term $-g^2$ of (11)); continuity then draws liquid towards the disk (the term $-2f'$ of the radial velocity and the axial velocity $v_z \propto -f$). With suction the axial flow is imposed at the disk ($f(0) = \lambda_1$). The velocity profile has the same form at every radius because v_r and v_θ scale with r: that is the self-similarity.
- **Limits.** For $\lambda_1 = 0$ (impermeable disk) the equations (4) and (11) with the conditions (5) are those of the rotating disk of the classical similarity solution; for $\Omega \to 0$ the whole flow vanishes. The pressure (13) is independent of r: the pressure of the problem varies only across the layer.
- **Dimensional check.** $\eta = (\Omega/\nu)^{1/2} z$ is dimensionless ($\sqrt{\mathrm{s^{-1}\,s/m^2}}\,\mathrm{m} = 1$). $(\nu\Omega)^{1/2}$ has units $\mathrm{m/s}$, so v_z is a velocity; $r\Omega$ is a velocity. $\mu\Omega$ has units $\mathrm{Pa\,s \cdot s^{-1}} = \mathrm{Pa}$, so (8) and (13) are pressures.
- **Consistency of (11).** Differentiating (8) with respect to r at fixed η shows that $\partial p/\partial r = C'(r)$ and (11) makes the radial momentum balance independent of r, as the similarity ansatz requires.